DICLOFENAC

Cardiovascular & GI risk

Avoid in 3rd trimester

CHEMICAL NAME 2-[2-(2,6-dichloroanilino)phenyl]acetic acid (as sodium or potassium salt)
CLASS Non-selective NSAID (acetic acid derivative)
TRADE NAMES Voltaren · Cataflam · Olfen · Diclac

THERAPEUTIC USES / INDICATIONS

- Musculoskeletal pain and inflammation: arthritis, back pain, sprains.
- Renal and biliary colic.
- Post-operative and dental pain; dysmenorrhoea.
- Acute gout (when colchicine unsuitable).

ADMINISTRATION

- **PO:** enteric-coated or modified-release tablets — swallow whole with food.
- **IM:** deep gluteal injection; max 2 days.
- **PR:** suppositories; **topical** gel.
- **IV:** only as diluted infusion in hospital, in hydrated patients.

DOSAGES

- **Adult PO:** 75–150 mg/day in 2–3 divided doses; max 150 mg/day.
- **IM:** 75 mg once (twice daily in severe cases) for max 2 days.
- **PR:** 50–100 mg up to 150 mg/day total from all routes.
- **Children (>1 yr, arthritis):** 1–3 mg/kg/day divided.

PHARMACOKINETICS

- **Onset:** PO 30–60 min; IM 15–30 min. **Duration:** 6–8 h.
- ~50% first-pass metabolism; highly protein-bound; metabolised in liver (CYP2C9).
- Excreted in urine and bile. **t½:** 1–2 h.

PHARMACODYNAMICS

- Inhibits **COX-1 and COX-2** (relatively COX-2 preferring) → ↓ prostaglandins.
- Analgesic, anti-inflammatory and antipyretic effects.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dyspepsia, nausea, diarrhoea, abdominal pain.
- Headache, dizziness.
- Injection-site pain; rash.

ADVERSE EFFECTS (SERIOUS)

- GI bleeding and ulceration.
- ↑ risk of MI and stroke (higher than ibuprofen).
- Acute kidney injury; hyperkalaemia.
- Hepatotoxicity; anaphylaxis; injection-site necrosis.

CONTRAINDICATIONS & CAUTIONS

- Ischaemic heart disease, stroke, peripheral arterial disease, heart failure (NYHA II–IV).
- Active peptic ulcer or GI bleeding.
- Severe renal or hepatic impairment.
- NSAID-sensitive asthma; third trimester of pregnancy.

KEY INTERACTIONS

- **Anticoagulants, antiplatelets, SSRIs:** ↑ bleeding.
- **ACE inhibitors/ARBs, diuretics:** ↓ effect, ↑ renal risk.
- **Lithium, methotrexate, ciclosporin:** ↑ toxicity.

NURSING MANAGEMENT

- Ask about heart disease, ulcers, asthma and kidney disease before the first dose.
- IM: deep upper-outer gluteal site; rotate sites; observe for local reactions.
- Monitor BP, urine output, signs of GI bleeding.
- Long-term: check renal function, LFTs and CBC.
- Assess pain response.

PATIENT EDUCATION

- Take tablets with food; swallow whole.
- Report chest pain, breathlessness, black stools or swelling.
- Do not combine with other NSAIDs or alcohol.
- Use the lowest dose for the shortest time.

KEY POINTS

- Strong anti-inflammatory NSAID.
- Max 150 mg/day, all routes combined.
- Higher CV risk — avoid in heart disease.
- Useful for renal colic.